

ASSIGNMENT #05

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2026
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Due Date: Friday, April 10th TA's Mailbox

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Abstract

Chemical engineering is about optimizing the performance of a production line. This is done through careful reaction engineering which includes property evaluations of the different components within the reactors. Since the processing of materials requires multiple components, the need to understand how properties change as result of mixing is of the utmost importance. Many considerations are needed to ensure safety of the reactor and the optimization of the process.

Properties of separate components within a mixture not only change significantly from their pure states but also as a function of the composition of the mixture. The property changes are a result from the interactions of the intermolecular forces on a microscopic level. Intermolecular forces change based on the interactions of the molecules. If the molecules are interacting with anything other than itself, there will be differences in the strength of the intermolecular forces. The rate that this changes with respect to the composition is known as Partial Molar Properties.

Partial molar properties calculate the different thermal properties of a mixture. This assignment covers the ideal case of partial molar properties treating the mixtures of vapor and liquid as ideal. This ideality demonstrates the importance of this concept in reactor and reaction design.

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1 Introduction

Experiments have shown that when separate components are brought together in a mixture, the pure molar thermal properties are not simply additive. For example, when 27.27 moles of methanol and 72.37 moles of water are mixed together, the additive volume is 2424.52 cm^3 . However, in actuality, the volume demonstrates a shrinkage of volume to 2344.70 cm^3 . Intermolecular forces differ in the mixture of methanol and water in comparison to the pure components on their own. Both polar components interact in such a way that the forces appear to hold more tightly resulting in the shrinkage of volume observed in experiments.

Mixtures also exhibit a temperature range during the 2-phase region between liquid and vapor. Pure components remain at the same temperature during the 2-phase region as the fluid heats from liquid. The following phase diagram of nitric acid and water show this range of temperatures based on the composition of the mixture.

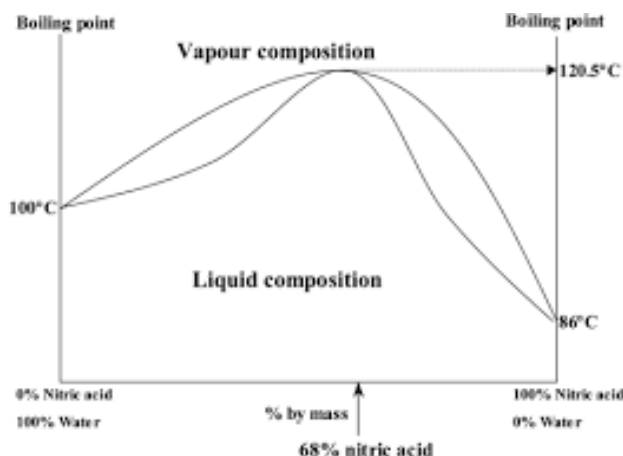


Figure 1: Vapor-Liquid Phase Diagram of Nitric Acid as a function of composition and temperature

The top boundary line is known as the dew line. Along this line the dew temperature (analogous to saturated vapor) can be identified for a certain composition. Along the bottom curve the bubble point temperature (analogous to saturated liquid) can be determined for a certain composition. At most compositions, there is a range of temperatures between the bubble point temperature and the dew point temperature. This phenomenon demonstrated in the volume example of methanol and water and the phase diagram of nitric acid and water, indicate that the intermolecular interactions affect the behavior of the components such that the contributions of each component differ from their pure molar thermal properties.

The difference in the pure thermal properties and the properties in mixtures is known as partial molar properties. Partial molar properties are dependent on the composition of the mixture indicating that a component of a mixture does not have the same partial molar properties in different mixtures. Intermolecular forces change based on the components and the composition of the mixture. Ultimately, so do the partial molar properties.

This assignment applies partial molar properties to two different scenarios. Problem A presents a isenthalpic throttling of a mixture that begins as a sub-cooled liquid under high pressure and ends as a 2-phase mixture. Dew and bubble point temperatures are used as reference points to ensure that the resulting answers exhibit common reasoning. Utilizing partial molar enthalpy ultimately solves for the composition vapor fraction and the liquid and vapor molar compositions. Rachford Rice's equation and Raoult's law are extensively used to also aid in the calculation.

Problem B invokes Henry's law for gases that are above their critical temperature and relative humidity to find the solubility of oxygen in pure water. Once again, partial molar pressures and Henry constants are used to find the concentration of oxygen in pure water. Since most of our world is a world of mixtures, ocean salt water, reaction in chemical reactors, and so on, it is crucial to understand and demonstrate the need for partial molar properties.

2 Problem A

2.1 Problem Statement

A mixture of ethylene, propane and iso-butane at temperature $T_1 = 0^\circ\text{C}$ and pressure $P_1 = 20$ bar is throttled isenthalpically down to a pressure $P_2 = 1$ bar as shown in the following figure:

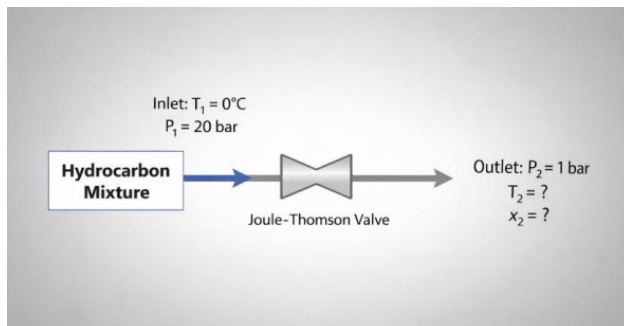


Figure 2: Diagram Representation of Problem A

The composition of the mixture is $z = [0.2, 0.5, 0.3]$ respectively. Assume that both the gas and liquid phases are ideal.

Assuming that both the gas and liquid are ideal allows for the calculations to make use of the ideal gas law and the ideal functions in the CHE calculator. Before calculations are addressed however, the mathematical definition of partial molar properties is needed.

2.1.1 Partial Molar Properties

The rate at which a molar thermal property changes with composition is given as:

$$\bar{M}_i = \left. \frac{\delta(nM)}{\delta n_i} \right|_{P, T, n_{j \neq i}}$$

The partial derivative indicates that as temperature, pressure, and all other molar compositions remain constant, the partial molar property ($\bar{M}$) of species i changes with respect to the molar composition of i . This is the mathematical support for the graphically demonstrated observation given in the introduction.

Two identities are developed through the use of the total differentiation of the partial molar property of the mixture and the relationship to the partial derivative of a partial molar property of a component. These are given as:

Constitutive Identity:

$$M = \sum \bar{M}_i x_i$$

Gibbs Thermodynamic Consistency Equation:

$$\sum x_i d\bar{M}_i = \frac{\delta M}{\delta P} \Big|_{T, n} dP + \frac{\delta M}{\delta T} \Big|_{P, n} dT$$

Under constant pressure and temperature, the Gibbs Thermodynamic Consistency Equation becomes:

$$\sum x_i d\bar{M}_i = 0$$

These two properties demonstrate two important characteristics of partial molar properties. The first from the constitutive property, is that the sum of the individual partial molar properties is equal to the molar thermal property of the mixture. The second, from Gibbs's Thermodynamic Consistency Equation is the the sum of the rate of the change in the partial molar properties multiplied by the respective molar composition must equal zero. For example, a binary system will see one species change increase while the

other decreases. The changes can vary in magnitude and sign depending on the mixture and the components. These properties lead to Rachford-Rice equation and the relationship for vapor-liquid equilibrium.

2.1.2 Rachford-Rice Equation

The Rachford-Rice equation was developed in 1952 by Rachford and Rice to help solve vapor-liquid equilibrium calculations. Before this equation was developed, the calculation required $2N+2$ equations for $2N+2$ unknowns for a system, where N is the number of components. For systems with a significant amount of components, these calculations are tedious at best. The Rachford-Rice equation uses Raoult's law and the lever rule so that only one unknown is required to solve the vapor-liquid equilibrium.

Raoult's law applies to ideal mixtures where the intermolecular forces are equal between similar components. It states that the vapor pressure of a solvent above a solution, or in the vapor phase in a vapor-liquid equilibrium, is equal to the saturated vapor pressure of the pure solvent at that temperature scaled by its mole fraction in the solution. This gives the following relationship where x_i is the liquid mole fraction:

$$P_i = x_i P_i^{sat}$$

This is just for the liquid phase of the equilibrium. The relation to the vapor phase is given by Dalton's Law of partial pressures. It states that the total pressure of a non reacting mixture of gases is equal to the sum of the partial pressures of each component in the mixture. Partial pressures are given as follows where y_i is the vapor mole fraction and P is the pressure of the system:

$$P_i = y_i P$$

These two laws can be brought together through the shared relationship for P_i in the following manner:

$$\begin{aligned} x_i P_i^{sat} &= y_i P \\ \frac{y_i}{x_i} &= \frac{P_i^{sat}}{P} = k_i \end{aligned}$$

This relationship gives rise to a new variable known as the "k-values". This k-value reduces the amount of unknowns from $2N$ to just N as it is dependent on both x_i and y_i .

The lever rule details the relationship of a 2-phase mixture using α as the vapor fraction. The vapor fraction determines the amount of vapor versus liquid in a mixture. The vapor fraction will be the same for each component in the mixture but the amounts of each are determined by the mole fractions. The lever rule is given as follows where F is the overall composition, z_i is the overall mole fraction of species i , V is the vapor composition, and L is the liquid composition:

$$\begin{aligned} F z_i &= V y_i + L x_i \\ \alpha &= \frac{V}{F} \\ z_i &= \alpha y_i + (1 - \alpha) x_i \end{aligned}$$

This lever rule is simplified using the k-value relationship, and then further solved for x_i and y_i :

$$\begin{aligned} z_i &= \alpha x_i k_i + (1 - \alpha) x_i \\ z_i &= x_i (1 + \alpha(k_i - 1)) \\ x_i &= \frac{z_i}{1 + \alpha(k_i - 1)} \\ y_i &= k_i x_i = \frac{z_i k_i}{1 + \alpha(k_i - 1)} \end{aligned}$$

Finally, we use the relationship that the sum of all liquid mole fractions and the sum of all vapor mole fractions both equal one. This allows for the relationship that derives the Rachford-Rice equation:

$$\begin{aligned}
\sum x_i &= 1 \\
\sum y_i &= 1 \\
\sum x_i - \sum y_i &= 0 \\
\sum \frac{z_i(k_i - 1)}{1 + \alpha(k_i - 1)} &= 0
\end{aligned}$$

The Rachford-Rice equation solves for the equilibrium phase split based on component feed mole fractions (z_i) which are typically known and the equilibrium ratios (k_i) which can be determined from the pressure of the system and the saturated vapor pressures of the components with respect to the temperature.

The implementation of the Rachford-Rice equation for Problem A will be discussed further in the calculations of dew temperature, bubble point temperature, and the solving of the temperature of the stream after throttling.

2.1.3 Partial Molar Enthalpy

Since this problem deals only with the enthalpy of the mixture, the remaining discussion of the partial molar properties will focus on developing the equations needed to calculate the enthalpy for each species and consequently the total mixture.

Just as with pure components, enthalpy is a function of temperature only such that for an ideal gas the enthalpy of a partial molar property of species i is as follows:

$$\bar{H}_i^{IG} = H_i^{IG}(P_i, T) = H_i^{IG}(P, T)$$

where P_i is the partial pressure of species i . Since enthalpy of an ideal gas is not pressure dependent, the partial pressure of species i is not needed; only the temperature of the mixture. This is calculated using the =HIG(compi, temp) function in the CHE calculator.

For an ideal liquid, the heat of vaporization is also needed to find the enthalpy of the liquid. Just as with the vapor, enthalpy is not dependent upon pressure and the equation is simply:

$$\bar{H}_i^{LD} = H_i = H_i^{IG}(P, T) - H_i^{vap}(P, T)$$

where $H_i^{vap}(P, T)$ is found from the CHE calculator =HVAP(compi, temp). The total enthalpy of the mixture is the sum of each species' enthalpies multiplied by the molar compositions through the constitutive identity.

$$H = \sum \bar{H}_i x_i$$

These equations and derivations allow for Problem A to be solved for a mixture. Even though partial molar properties are needed, the same basic principles of thermodynamics apply to the throttling process. This is demonstrated and discussed in the following sections.

2.2 Bubble Point and Dew Point Temperature of the Mixture

Bubble point and dew point temperatures were briefly described in the introduction. Bubble point is associated with the saturated liquid point because it is imagined that first boiling bubble that appears is when the mixture enters the 2-phase region from a sub-cooled liquid. This allows for the vapor fraction to be set to equal 0. Conversely, dew temperature is associated with the saturated vapor point because it is imagined as the first droplet of the mixture to start condensing into the 2-phase region from a superheated vapor. This allows for the vapor fraction to be set equal to 1.

When $\alpha=1$, the Rachford-Rice equation simplifies to the following:

$$\sum \frac{z_i(k_i - 1)}{1 + (1)(k_i - 1)} = 0$$

$$\sum \frac{z_i(k_i - 1)}{k_i} = 0$$

$$\sum \left(\frac{z_i k_i}{k_i} \right) - \sum \frac{z_i}{k_i} = 0$$

$$\sum z_i - \sum \frac{z_i}{k_i} = 0$$

$$1 - \sum \frac{z_i}{k_i} = 0$$

$$1 = \sum \frac{z_i}{k_i}$$

At the dew point when $\alpha=1$, we find that the sum of the overall mole fractions divided by the k-values of each species will equal one. To calculate the dew point temperature, the excel sheet was set up with an initial guess for the temperature, 90 °C. This guess was based on the saturated temperature values of each component at 20 bar (iso-butane). The Dew Point temperature will be less than the largest saturated temperature, but will still be effected by it. Using the saturated vapor pressure function in the CHE calculator, =Psat(compi,temp), each components saturated vapor pressure was found for the initial guess's temperature. The k-values were then calculated using the pressure relationship rather than the mole fraction relationship:

$$k_i = \frac{P_i^{sat}}{P}$$

The Rachford-Rice equation was then used to find the $\frac{z_i}{k_i}$ for each component. After each component was found, the summation was taken. Goal seek was then used to set the summation equal to 1 by changing the temperature. The resulting temperature was found to be 72.8 °C. The following figures show the set up of the excel, the answer, and then the formulas used in each cell:

	A	B	C	D
1		Stream 1		
2		Ethylene	Propane	iso-Butane
3	Temp, C	0		
4	Pressure, bar	20		
5	zi	0.2	0.5	0.3
6	P,sati	36.46389238	4.611806446	1.567865903
7	ki	1.823194619	0.230590322	0.078393295
8	T,sati	-26.37613654	62.27441197	107.1550891
15	Dew Point Calculations			
16	T,dew oC	72.80305027		
17	P,sati	119.4497935	24.16693732	10.85531234
18	ki	5.972489677	1.208346866	0.542765617
19	zi/ki	0.033486872	0.413788469	0.552724768
20	sum of row 17	1.00000011		

Figure 3: Excel Setup for Dew Point Temperature Calculation

	A	B	C	D
1			Stream 1	
2		Ethylene	Propane	iso-Butane
3	Temp, C	0		
4	Pressure, bar	20		
5	zi	0.2	0.5	0.3
6	P,sati	=Psat(comp1,B3)	=Psat(comp2,B3)	=Psat(comp3,B3)
7	ki	=B6/\$B\$4	=C6/\$B\$4	=D6/\$B\$4
8	T,sati	=Tsat(comp1,B4)	=Tsat(comp2,B4)	=Tsat(comp3,B4)
15		Dew Point Calculations		
16	T,dew oC	72.8030502657153		
17	P,sati	=Psat(comp1,B16)	=Psat(comp2,B16)	=Psat(comp3,B16)
18	ki	=B17/\$B\$4	=C17/\$B\$4	=D17/\$B\$4
19	zi/ki	=B5/B18	=C5/C18	=D5/D18
20	sum of row 17	=SUM(B19:D19)		

Figure 4: Formula Audit of the Dew Point Temperature Calculation

The same process in excel was used to find the bubble point temperature. The derivation from the Rachford-Rice equation is different however. Bubble point temperature is analogous to the saturate liquid point when $\alpha=0$. The Rachford-Rice equation derivation is as follows:

$$\sum \frac{z_i(k_i - 1)}{1 + (0)(k_i - 1)} = 0$$

$$\sum \frac{z_i(k_i - 1)}{1} = 0$$

$$\sum z_i k_i - \sum z_i = 0$$

$$\sum z_i k_i - 1 = 0$$

$$1 = \sum z_i k_i$$

The following figures are for the excel set up of the bubble point temperature calculation and the formula audit of excel. The initial guess for the bubble point was -20°C guessed from the smallest saturated temperature from the components (ethylene). The bubble point temperature of the mixture before throttling is found to be 32.7°C .

	A	B	C	D
1			Stream 1	
2		Ethylene	Propane	iso-Butane
3	Temp, C		0	
4	Pressure, bar		20	
5	zi	0.2	0.5	0.3
6	P,sati	36.46389238	4.611806	1.567866
7	ki	1.823194619	0.23059	0.078393
8	T,sati	-26.3761365	62.27441	107.1551
9		Bubble Point Calculations		
10	T,bub oC		32.74914744	
11	P,sati	66.39675639	10.86489	4.294011
12	ki	3.319837819	0.543245	0.214701
13	zi*Ki	0.663967564	0.271622	0.06441
14	Sum of row 12		0.999999995	

Figure 5: Excel Setup for Bubble Point Temperature Calculation

	A	B	C	D
1			Stream 1	
2		Ethylene	Propane	iso-Butane
3	Temp, C	0		
4	Pressure, bar	20		
5	zi	0.2	0.5	0.3
6	P,sati	=Psat(comp1,B3)	=Psat(comp2,B3)	=Psat(comp3,B3)
7	ki	=B6/\$B\$4	=C6/\$B\$4	=D6/\$B\$4
8	T,sati	=Tsat(comp1,B4)	=Tsat(comp2,B4)	=Tsat(comp3,B4)
9	Bubble Point Calculations			
10	T,bub oC	32.749147439654		
11	P,sati	=Psat(comp1,B10)	=Psat(comp2,B10)	=Psat(comp3,B10)
12	ki	=B11/B4	=C11/B4	=D11/B4
13	zi*Ki	=B5*B12	=C5*C12	=D5*D12
14	Sum of row 12	=SUM(B13:D13)		
15	Dew Point Calculations			
16	T,dew oC	72.8030502657153		
17	P,sati	=Psat(comp1,B16)	=Psat(comp2,B16)	=Psat(comp3,B16)
18	ki	=B17/\$B\$4	=C17/\$B\$4	=D17/\$B\$4
19	zi/ki	=B5/B18	=C5/C18	=D5/D18
20	sum of row 17	=SUM(B19:D19)		

Figure 6: Formula Audit of the Bubble Point Temperature Calculation

2.3 Temperature of the Outlet Valve and Vapor Fraction

Throttling is an isenthalpic process meaning that the enthalpy does not change from stream 1 to stream 2. Therefore, this calculation first needs the enthalpy of stream 1. Since the bubble point temperature is 32.7°C , stream 1 at 20 bar and 0°C is determined to be a sub-cooled liquid. Therefore, the enthalpy is calculated using the constitutive identity discussed in the introduction.

$$H = \sum x_i \bar{H}_i$$

$$\bar{H}_i = H_i^{IG} - H_i^{vap}$$

The following figures depict the excel set up and formulas used to calculate the individual partial molar enthalpies and the overall enthalpy of the mixture.

	A	B	C	D
1			Stream 1	
2		Ethylene	Propane	iso-Butane
3	Temp, C		0	
4	Pressure, bar		20	
5	zi	0.2	0.5	0.3
6	P,sati	36.46389238	4.611806446	1.567865903
7	ki	1.823194619	0.230590322	0.078393295
8	T,sati	-26.37613654	62.27441197	107.1550891
21	Enthalpy Calculations			
22	Hi, kJ/(kmol*k)	45979.99179	-122099.1177	-157607.085
23	zi*Hi	9195.998358	-61049.55886	-47282.12549
24	H, kJ/(kmol*K)		-99135.68599	

Figure 7: Enthalpy Calculation of Stream 1

	A	B	C	D
1			Stream 1	
2		Ethylene	Propane	iso-Butane
3	Temp, C	0		
4	Pressure, bar	20		
5	zi	0.2	0.5	0.3
6	P,sat	=Psat(comp1,B3)	=Psat(comp2,B3)	=Psat(comp3,B3)
7	ki	=B6/\$B\$4	=C6/\$B\$4	=D6/\$B\$4
8	T,sat	=Tsat(comp1,B4)	=Tsat(comp2,B4)	=Tsat(comp3,B4)
21	Enthalpy Calculations			
22	Hi, kJ/(kmol*k)	=HIG(comp1,B3)-Hvap(comp1,B3)	=HIG(comp2,B3)-Hvap(comp2,B3)	=HIG(comp3,B3)-Hvap(comp3,B3)
23	zi*Hi	=B22*B5	=C22*C5	=D22*D5
24	H, kJ/(kmol*K)	=SUM(B23:D23)		

Figure 8: Formula Audit of the Enthalpy Calculation of Stream 1

Before enthalpy for stream two can be calculated, the phase of the mixture needs to be identified. This is done by setting α to one and zero and find the value of the summation for each. The mixture is in the two phase region when solving the Rachford-Rice equation at $f(\alpha=0)>0$ and $f(\alpha=1)<0$. The monotonic nature of the Rachford-Rice equation between 0 and 1 for the vapor fraction is what allows for this observation. The following figures show the values calculated in excel and the formulas used.

	F	G	H	I	J	K	L
1			Stream 2				
2		Ethylene	Propane	iso-Butane			
3	Temp, oC		-45.69757651			Rachford-Rice to determine 2-phase status	
4	Press, bar		1			alpha	Rachford-Rice Summation
5	zi	0.2	0.5	0.3		1	-1.018498144
6	Psati	11.71466145	0.853062555	0.211968889		0	1.833054235
7	ki	11.71466145	0.853062555	0.211968889			

Figure 9: Calculation of $f(\alpha=0)$ and $f(\alpha=1)$ to determine phase

	F	G	H	I	J	K	L
1			Stream 2				
2		Ethylene	Propane	iso-Butane			
3	Temp, oC	-45.697576509587					Rachford-Rice to determine 2-phase status
4	Press, bar	1				alpha	Rachford-Rice Summation
5	zi	0.2	0.5	0.3		1	=1-G5/G7+H5/H7-I5/I7
6	Psati	=Psat(comp1,G3)	=Psat(comp2,G3)	=Psat(comp3,G3)		0	=SUMPRODUCT(G5:I5,G7:I7)-1
7	ki	=G6/G\$4	=H6/G\$4	=I6/G\$4			

Figure 10: Formula Audit of Phase Determination Values

Confirming that the mixture is in the two phase region, bubble point temperature and dew point temperature are needed to know the bounds of the temperature of stream two. These calculations were approached using the same derivation and methods completed in stream one. The following figures provide the dew point and bubble point temperature calculations and values, and the excel formulas used to calculate. Goal Seek was utilized as the numerical method to find the value of the temperatures by setting the vapor fraction equal to one. The initial temperature guess for dew point was $-20\text{ }^{\circ}\text{C}$ and for bubble point $-50\text{ }^{\circ}\text{C}$ based on the saturated vapor temperatures for the components. Results show that the mixture in the two phase vapor region at 1 bar must be a temperature between $-31.0\text{ }^{\circ}\text{C}$ and $-73.4\text{ }^{\circ}\text{C}$.

	F	G	H	I
2		Ethylene	Propane	iso-Butane
3	Temp, oC	-45.69757651		
4	Press, bar	1		
5	zi	0.2	0.5	0.3
6	Psati	11.71466145	0.853062555	0.211968889
7	ki	11.71466145	0.853062555	0.211968889
8	Tsat,i	-103.9657298	-42.15250313	-12.1014586
12	Bubble Point Calculations			
13	T, bub oC	-73.36634088		
14	P,sati	4.453554953	0.196405891	0.03695355
15	ki	4.453554953	0.196405891	0.03695355
16	zi*Ki	0.890710991	0.098202945	0.011086065
17	Sum of row 15	1.000000001		
18	Dew Point Calculations			
19	T,dew oC	-31.00231227		
20	P,sati	17.74447786	1.591679762	0.444711043
21	ki	17.74447786	1.591679762	0.444711043
22	zi/ki	0.011271112	0.314133541	0.674595347
23	sum of row 17	1		

Figure 11: Dew Point and Bubble Point Temperatures of the Mixture in Stream 2

	F	G	H	I
1			Stream 2	
2		Ethylene	Propane	iso-Butane
3	Temp, oC	-45.697576509587		
4	Press, bar	1		
5	zi	0.2	0.5	0.3
6	Psati	=Psat(comp1,G3)	=Psat(comp2,G3)	=Psat(comp3,G3)
7	ki	=G6/\$G\$4	=H6/\$G\$4	=I6/\$G\$4
8	Tsat,i	=Tsat(comp1,G4)	=Tsat(comp2,G4)	=Tsat(comp3,G4)
12	Bubble Point Calculations			
13	T, bub oC	-73.3663408800173		
14	P,sati	=Psat(comp1,G13)	=Psat(comp2,G13)	=Psat(comp3,G13)
15	ki	=G14/\$G\$4	=H14/\$G\$4	=I14/\$G\$4
16	zi*Ki	=G15*G5	=H15*H5	=I15*I5
17	Sum of row 15	=SUM(G16:I16)		
18	Dew Point Calculations			
19	T,dew oC	-31.0023122664613		
20	P,sati	=Psat(comp1,G19)	=Psat(comp2,G19)	=Psat(comp3,G19)
21	ki	=G20/\$G\$4	=H20/\$G\$4	=I20/\$G\$4
22	zi/ki	=G5/G21	=H5/H21	=I5/I21
23	sum of row 17	=SUM(G22:I22)		

Figure 12: Formula Audit of the Dew Point and Bubble Point Temperature Calculations

With the temperature boundaries known, the vapor fraction and the resulting temperature can be calculated. Multiple ways of solving for the vapor fraction which include solving for α explicitly or using solver with constraints. The vapor fraction here was solved explicitly but also connecting the terms from the developed quadratic formula so that the temperature change also changed the values.

To solve for the vapor fraction explicitly, the Rachford-Rice equation is solved for α symbolically first:

$$\sum \frac{z_i(k_i - 1)}{1 + \alpha(k_i - 1)} = \frac{z_1(k_1 - 1)}{1 + \alpha(k_1 - 1)} + \frac{z_2(k_2 - 1)}{1 + \alpha(k_2 - 1)} + \frac{z_3(k_3 - 1)}{1 + \alpha(k_3 - 1)} = 0$$

$$z_1(k_1 - 1)(1 + \alpha(k_2 - 1))(1 + \alpha(k_3 - 1)) + z_2(k_2 - 1)(1 + \alpha(k_1 - 1))(1 + \alpha(k_3 - 1)) + z_3(k_3 - 1)(1 + \alpha(k_2 - 1))(1 + \alpha(k_1 - 1)) = 0$$

The quadratic formula derived from the previous formula gives the following coefficients for α :

$$a = (k_1 - 1)(k_2 - 1)(k_3 - 1)$$

$$b = z_1(k_1 - 1)(k_2 + k_3 - 2) + z_2(k_2 - 1)(k_1 + k_3 - 2) + z_3(k_3 - 1)(k_1 + k_2 - 2)$$

$$c = z_1k_1 + z_2k_2 + z_3k_3$$

The coefficients were calculated so that as the temperature and alpha changed, the liquid and vapor mole fractions would also change. The temperature changes as a result of the enthalpy balance. As previously mentioned, the enthalpy between the two streams must be equal. Therefore, to calculate the enthalpy of stream two, both vapor and liquid enthalpies of each component must be calculated and the lever rule must be applied. The enthalpy for the vapor and liquid enthalpies were calculated as previously derived in the introduction. The total molar enthalpy for both vapor and liquid are scaled by their respective mole fractions and summed. The lever rule is then applied so that the overall enthalpy of the mixture is calculated as follows:

$$H = H^V \alpha - (1 - \alpha)H^L$$

$$H^V = \sum y_i H_i^{\bar{G}}$$

$$H^L = \sum x_i (H_i^{\bar{G}} - H_i^{\bar{vap}})$$

Excel was set up with initial guess of 0.5 for the vapor fraction with an enthalpy balance. The enthalpy balance was goal seeked to a value of zero by changing the temperature. The resulting vapor fraction and temperature were calculated to be 0.3857 and -45.7 °C respectively. The following figures show the full excel set up and the formula audit.

	A	B	C	D	E	F	G	H	I	J	K	L
1			Stream 1				Stream 2				Quadratic Formula Coefficients	
2		Ethylene	Propane	iso-Butane			Ethylene	Propane	iso-Butane		a	1.240664343
3	Temp, C	0	0			Temp, oC	-45.69757651				b	-5.231179745
4	Pressure, bar	20	20			Press, bar	1				c	1.833054235
5	zi	0.2	0.5	0.3		zi	0.2	0.5	0.3			
6	P.sati	36.46389238	4.611806446	1.567865903		Psati	11.71466145	0.853062555	0.211968889			
7	ki	1.823194619	0.230590322	0.078393295		ki	11.71466145	0.853062555	0.211968889			
8	T.sati	-26.37613654	62.27441197	107.1550891		Tsat,i	-103.9657298	-42.15250313	-12.1014586			
9	Bubble Point Calculations					alpha	0.385689473					
10	T.bub oC	32.74914744				xi	0.038967121	0.530038459	0.43099442	1	Rachford-Rice to determine 2-phase status	
11	P.sati	66.39675639	10.86489088	4.29401058		yi	0.456486629	0.452155962	0.091357408	1		
12	ki	3.319837819	0.543244544	0.214700529		Bubble Point Calculations					alpha	Rachford-Rice Summation
13	zi*ki	0.663967564	0.271622272	0.064410159		T. bub oC	-73.36634088				1	-1.018498144
14	Sum of row 12	0.999999995				P.sati	4.453554953	0.196405891	0.03695355		0	1.833054235
15	Dew Point Calculations					ki	4.453554953	0.196405891	0.03695355			
16	T.dew oC	72.80305027				zi*ki	0.890710991	0.098202945	0.011086065			
17	P.sati	119.4497935	24.16693732	10.85531234		Sum of row 15	1.000000001					
18	ki	5.972489677	1.208346866	0.542765617		Dew Point Calculations						
19	zi/ki	0.033486872	0.413788469	0.552724768		T.dew oC	-31.00231227					
20	sum of row 17	1.000000011				P.sati	17.74447786	1.591679762	0.444711043			
21	Enthalpy Calculations					ki	17.74447786	1.591679762	0.444711043			
22	Hi, kJ/(kmol*K)	45979.99179	-122099.1177	-157607.085		zi/ki	0.011271112	0.314133541	0.674595347			
23	zi*Hi	9195.998358	-61049.55886	-47282.12549		sum of row 17	1					
24	H, kJ/(kmol*K)	-99135.68599				Enthalpy Calculations						
25						Hi,I	39201.89736	-127540.6091	-163821.4573			
26	Enthalpy Balance:	-5.79865E-07				xi*Hi	1527.585075	-67601.42792	-70606.13397			
27						H,I		-136679.9768				
28						Hi,v	49543.81803	-108570.672	-140786.3745			
29						yi*Hi,v	22616.09049	-49090.87664	-12861.87832			
30						H,v		-39336.66447				
31						H,tot		-99135.68599				

Figure 13: Excel Sheet for Problem A with all Calculated Values

	A	B	C	D	E	F	G	H	I	J	K	L
1			Stream 1				Stream 2				Quadratic Formula Coefficients	
2		Ethylene	Propane	iso-Butane			Ethylene	Propane	iso-Butane		a	=(G7-1)*(H7-1)*(I7-1)
3	Temp, C	0	0			Temp, oC	-45.697576509587				b	=G5*(G7-1)*(H7-1)*(I7-1)+G7-2*(H7-1)*(I7-1)+G7-2
4	Pressure, bar	20	20			Press, bar	1				c	=G5*(G7-1)*(H7-1)*(I7-1)
5	zi	0.2	0.5	0.3		zi	0.2	0.5	0.3			
6	P.sati	=Psat(comp1,B3)	=Psat(comp2,B3)	=Psat(comp3,B3)		Psati	=Psat(comp1,G3)	=Psat(comp2,G3)	=Psat(comp3,G3)			
7	ki	=K1(B5:B84)	=K1(B5:B84)	=K1(B5:B84)		ki	=K1(G5:G84)	=K1(G5:G84)	=K1(G5:G84)			
8	T.sati	=Tsati(comp1,B4)	=Tsati(comp2,B4)	=Tsati(comp3,B4)		Tsati	=Tsati(comp1,G4)	=Tsati(comp2,G4)	=Tsati(comp3,G4)			
9	Bubble Point Calculations					alpha	=((1-500*(1-3+2*(1-2141)))/(2141))					
10	T.bub oC	32.749147439554	=Psat(comp2,B10)	=Psat(comp3,B10)		xi	=G5*(1-500*(H7-1))	=H5*(1-500*(I7-1))	=I5*(1-500*(J7-1))	=SUM(G10:I10)	Rachford-Rice to determine 2-phase status	
11	P.sati	=Psat(comp1,B10)	=Psat(comp2,B10)	=Psat(comp3,B10)		yi	=G10*(1-500*(H7-1))	=H10*(1-500*(I7-1))	=I10*(1-500*(J7-1))	=SUM(G11:I11)		
12	ki	=K1(B11)	=K1(B11)	=K1(B11)		Bubble Point Calculations					alpha	Rachford-Rice Summation
13	zi*ki	=G11*(K1(B11))	=G11*(K1(B11))	=G11*(K1(B11))		T.bub oC	=-73.366340880173	=Psat(comp2,G13)	=Psat(comp3,G13)		1	=1-G5*(G7-1)*(H7-1)*(I7-1)+G7-2*(H7-1)*(I7-1)+G7-2
14	Sum of row 12	=SUM(B13:D13)				P.sati	=Psat(comp1,G13)	=Psat(comp2,G13)	=Psat(comp3,G13)		0	=SUM(F10:J10)/(G5:I5,G7:I7)-1
15	Dew Point Calculations					ki	=K1(G14)	=K1(G14)	=K1(G14)			
16	T.dew oC	72.8030502657153	=Psat(comp2,B16)	=Psat(comp3,B16)		zi*ki	=G14*(K1(G14))	=G14*(K1(G14))	=G14*(K1(G14))			
17	P.sati	=Psat(comp1,B16)	=Psat(comp2,B16)	=Psat(comp3,B16)		Sum of row 15	=SUM(G15:I15)					
18	ki	=K1(B17)	=K1(B17)	=K1(B17)		T.dew oC	=-31.002312264613	=Psat(comp2,G19)	=Psat(comp3,G19)			
19	zi/ki	=G17*(K1(B17))	=G17*(K1(B17))	=G17*(K1(B17))		P.sati	=Psat(comp1,G19)	=Psat(comp2,G19)	=Psat(comp3,G19)			
20	sum of row 17	=SUM(B19:D19)				ki	=K1(G20)	=K1(G20)	=K1(G20)			
21	Enthalpy Calculations					zi/ki	=G20*(K1(G20))	=G20*(K1(G20))	=G20*(K1(G20))			
22	Hi, kJ/(kmol*K)	=H1(G21)	=H1(G21)	=H1(G21)		sum of row 17	=SUM(G22:I22)					
23	zi*Hi	=G21*(H1(G21))	=G21*(H1(G21))	=G21*(H1(G21))		Enthalpy Calculations						
24	H, kJ/(kmol*K)	=SUM(B23:D23)				Hi,I	=H1(G23)	=H1(G23)	=H1(G23)			
25						xi*Hi	=G23*(H1(G23))	=G23*(H1(G23))	=G23*(H1(G23))			
26	Enthalpy Balance:	-624-G31				H,I	=SUM(G24:I24)					
27						Hi,v	=H1(G25)	=H1(G25)	=H1(G25)			
28						yi*Hi,v	=G25*(H1(G25))	=G25*(H1(G25))	=G25*(H1(G25))			
29						H,v	=SUM(G26:I26)					
30						H,tot	=SUM(G27:I27)					

Figure 14: Formula Audit of Excel Sheet for Problem A

2.4 Problem A Conclusion

Up to this point, throttling and other thermodynamic processes have been evaluated for pure substances only. Assignments 1 and 2 evaluated streams of pure water through a Brayton and regenerative Rankine power cycle respectively. Evaluations of the streams in both power cycles were completed using energy balances and efficiency balances to account for the thermodynamic changes as result of the processing of the streams through different equipments. This assignment demonstrates that the thermodynamic principles for evaluating streams through equipment remains the same for a mixture, but the development of the thermal properties differs because of the partial molar properties of mixtures. Mixtures can provide essential thermal properties for making a process more efficient or just exist as a result of a reaction. Either way, understanding how the thermal properties change within a mixture is crucial to understand how the reactors or a system will behave.

3 Problem B

3.1 Problem Statement

Calculate the solubility of oxygen in water at $T = 25^\circ\text{C}$ and $P = 1\text{ atm}$. Assume that air behaves as an ideal gas and the liquid water solution is also ideal. Calculate the milligrams of oxygen per liter of pure water (mg/L) for this system. Assume that composition of dry air is 21% oxygen and 79% nitrogen.

3.1.1 Henry's Law

When gases are present in a mixture at a temperature above their critical temperature, Henry's law is used to describe the nature of the intermolecular forces within the system. Henry's law states that the amount of dissolved gas in a liquid is directly proportional to its partial pressure above that liquid at a constant temperature. Essentially, the higher the gas pressure becomes, the higher the solubility in the liquid phase. The correlation of the solubility is given in the following mathematical relationship, where H_i here is Henry's Constant of species i .

$$y_i P = H_i x_i$$
$$\frac{y_i}{x_i} = \frac{H_i}{P} = k_i$$

3.1.2 Relative Humidity

Relative humidity is a measurement of the actual amount of water vapor present in the air compared to the amount of water vapor the air could hold at a certain temperature. The calculation of relative humidity is the ratio of the partial pressure of water vapor to the saturated vapor pressure of water at that temperature:

$$RH = \frac{P_{H_2O}(T)}{P_{H_2O}^{sat}(T)}$$

3.2 Partial Pressure Calculations

3.2.1 Partial Pressure of Water

In the CHE calculator, the saturated vapor pressure of water was calculated using the function =WaterP-sat(temp). This found the saturated vapor pressure of water to be 0.03166 bar. The relationship between relative humidity and the saturated vapor pressure of water was invoked to determine that the partial pressure of water is 0.01583 bar.

3.2.2 Partial Pressure of Nitrogen and Oxygen

The partial pressure of water was subtracted from the total pressure of the system. The pressure given was in units of atm but was converted to bar. Therefore, the total pressure of the system in bar is 1.01325 and the pressure of nitrogen and oxygen is the difference between the total pressure and the partial pressure of water. The partial pressures of oxygen and nitrogen were calculated applying the mole fraction of dry air:

$$P_{N_2} = 0.79(P - P_{H_2O}) = 0.7880\text{bar}$$

$$P_{O_2} = 0.21(P - P_{H_2O}) = 0.2067\text{bar}$$

3.2.3 Mole Fraction of the Overall Composition, z_i

The mole fraction of the overall composition, z_i , is the ratio of the partial pressure of the component to the overall pressure of the ratio given by Dalton's Law of Partial Pressures.

$$z_i = \frac{P_i}{P}$$

The following figures provide the results of the calculations to this point and the formulas in excel used to calculate them.

	A	B	C	D
1		Water	Nitrogen	Oxygen
2	Temperature, oC		25	
3	Temperature, K		298.15	
4	Pressure, Bar		1.01325	
5	Psat, i, bar	0.03166	1323.499518	1761.229951
6	Tsat, i, oC	99.99983	-195.6875406	-182.8946437
7	Relative Humidity		0.5	
8	Pi, bar	0.01583	0.787961796	0.209458199
9	zi	0.015623	0.77765783	0.20671917

Figure 15: Initial Problem B Values

	A	B	C	D
1		Water	Nitrogen	Oxygen
2	Temperature, oC	25		
3	Temperature, K	=ctok(B2)		
4	Pressure, Bar	=ATMtoBAR(1)		
5	Psat, i, bar	=WaterPsat(B2)	=Psat(comp6,B2)	=Psat(comp4,B2)
6	Tsat, i, oC	=WaterTsat(B4)	=Tsat(comp6,B4)	=Tsat(comp4,B4)
7	Relative Humidity	0.5		
8	Pi, bar	=B5*B7	=0.79*(B4-B8)	=0.21*(B4-B8)
9	zi	=B8/\$B\$4	=C8/\$B\$4	=D8/\$B\$4

Figure 16: Formula Audit of the Initial Values for Problem B

3.2.4 k-Value Calculations

Under Raoult's law, the k-values were calculated using the ratio of the component's saturated vapor pressures to the pressure of the system. Henry's Law utilizes the Henry Constant. Henry's Constant is calculated using coefficient that correspond to specific components at specific temperatures using the following expression:

$$\ln H = A + \frac{B}{T} + C \ln T + DT$$

In the CHE calculator, the function =Henry(temp, coefficients) is used to calculate the constant. The coefficients are listed in their own sheet by component and what it is dissolving into. The following figure shows the table referenced in the excel sheet.

	A	B	C
1	Properties		
2	Henry Constants, bar	O2	N2
3		H2O	H2O
4	A	144.4081	164.9941
5	B	-7775.0600	-8432.7700
6	C	-18.3974	-21.5580
7	D	-0.0094	-0.0084

Figure 17: The Coefficients for Henry's Constant of Oxygen and Nitrogen dissolved into Water

The constant represents the amount of pressure required to "force" a specific mole fraction into the liquid. In this case, the Henry Constants are specifically for nitrogen's solubility into water and oxygen's solubility into water. Thus the units of Henry's Constant is bar.

The ratio of Henry's constant to the pressure of the system provides the k-value for nitrogen and oxygen only. The k value of water is calculated using Raoult's law as there is not a Henry Constant for water into water.

	A	B	C	D
1		Water	Nitrogen	Oxygen
2	Temperature, oC		25	
3	Temperature, K		298.15	
4	Pressure, Bar		1.01325	
5	Psat, i, bar	0.031660011	1323.499518	1761.229951
6	Tsat, I, oC	99.99983306	-195.6875406	-182.8946437
7	Relative Humidity		0.5	
8	Pi, bar	0.015830005	0.787961796	0.209458199
9	zi	0.015623001	0.77765783	0.20671917
10	Henry's Constants, Hi, bar		86367.81552	44083.38413
11	ki, bar	0.031246001	85238.40664	43506.91747

Figure 18: Values calculated for the Henry's Constants and the k-values

	A	B	C	D
1		Water	Nitrogen	Oxygen
2	Temperature, oC	25		
3	Temperature, K	=ctok(B2)		
4	Pressure, Bar	=ATMtoBAR(1)		
5	Psat, i, bar	=WaterPsat(B2)	=Psat(comp6,B2)	=Psat(comp4,B2)
6	Tsat, I, oC	=WaterTsat(B4)	=Tsat(comp6,B4)	=Tsat(comp4,B4)
7	Relative Humidity	0.5		
8	Pi, bar	=B5*B7	=0.79*(B4-B8)	=0.21*(B4-B8)
9	zi	=B8/\$B\$4	=C8/\$B\$4	=D8/\$B\$4
10	Henry's Constants, Hi, bar		=Henry(B2,'Henry Constants'!C4:C7)	=Henry(B2,'Henry Constants'!B4:B7)

Figure 19: Formula Audit of the Calculation of Henry's Constants and K-Values

3.2.5 Vapor Fraction Calculation

With the z_i and k_i values calculated, the vapor fraction is calculated to further support what the relative humidity tells us about the phase. The vapor fraction was calculated using a goal seek by setting the sum of each component's Rachford-Rice term to equal zero by changing the value of the super vapor fraction. The value of the vapor fraction is $\alpha=1.016$. This value indicates that the system is a superheated vapor. Relative humidity values below one also indicate that the vapor is superheated.

A superheated vapor indicates that the solubility of oxygen within pure water will be extremely small.

	A	B	C	D
1		Water	Nitrogen	Oxygen
2	Temperature, oC		25	
3	Temperature, K		298.15	
4	Pressure, Bar		1.01325	
5	Psat, i, bar	0.031660011	1323.499518	1761.229951
6	Tsat, I, oC	99.99983306	-195.6875406	-182.8946437
7	Relative Humidity		0.5	
8	Pi, bar	0.015830005	0.787961796	0.209458199
9	zi	0.015623001	0.77765783	0.20671917
10	Henry's Constants, Hi, bar		86367.81552	44083.38413
11	ki, bar	0.031246001	85238.40664	43506.91747
18	alpha		1.016126682	
19	Rachford-Rice	-0.968740771	0.765306989	0.203433782
20	Sum of RR		-6.12943E-10	

Figure 20: Vapor Fraction Calculation

	A	B	C	D
1		Water	Nitrogen	Oxygen
2	Temperature, oC	25		
3	Temperature, K	=ctok(B2)		
4	Pressure, Bar	=ATMtoBAR(1)		
5	Psat, i, bar	=WaterPsat(B2)	=Psat(comp6,B2)	=Psat(comp4,B2)
6	Tsat, I, oC	=WaterTsat(B4)	=Tsat(comp6,B4)	=Tsat(comp4,B4)
7	Relative Humidity	0.5		
8	Pi, bar	=B5*B7	=0.79*(B4-B8)	=0.21*(B4-B8)
9	zi	=B8/\$B\$4	=C8/\$B\$4	=D8/\$B\$4
10	Henry's Constants, Hi, bar		=Henry(B2,'Henry Constants'!C4:C7)	=Henry(B2,'Henry Constants'!B4:B7)
11	ki, bar	=B5/B4	=C10/B4	=D10/B4
18	alpha	1.01612668161347		
19	Rachford-Rice	=(B9*(B11-1))/(1+\$B\$18*(B11-1))	=(C9*(C11-1))/(1+\$B\$18*(C11-1))	=(D9*(D11-1))/(1+\$B\$18*(D11-1))
20	Sum of RR	=SUM(B19:D19)		

Figure 21: Formula Audit of the Vapor Fraction Calculation

3.2.6 Liquid Mole Fractions

Liquid mole fractions for nitrogen and oxygen followed Henry's Law through the following relationship:

$$x_i = \frac{z_i}{H_i}$$

The liquid mole fraction of water was calculated from the following relationship:

$$\sum x_i = 1$$

$$1 - x_{O_2} - x_{N_2} = x_{H_2O}$$

	A	B	C	D
1		Water	Nitrogen	Oxygen
2	Temperature,		25	
3	Temperature,		298.15	
4	Pressure, Bar		1.01325	
5	Psat, i, bar	0.031660011	1323.499518	1761.229951
6	Tsat, I, oC	99.99983306	-195.6875406	-182.8946437
7	Relative Hum		0.5	
8	Pi, bar	0.015830005	0.787961796	0.209458199
9	zi	0.015623001	0.77765783	0.20671917
10	Henry's Constants, Hi, bar		86367.81552	44083.38413
11	ki, bar	0.031246001	85238.40664	43506.91747
12	xi	0.999986125	9.12333E-06	4.75141E-06

Figure 22: Liquid Mole Fraction Results

	A	B	C	D
1		Water	Nitrogen	Oxygen
2	Temperature, oC	25		
3	Temperature, K	=ctok(B2)		
4	Pressure, Bar	=ATMtoBAR(1)		
5	Psat, i, bar	=WaterPsat(B2)	=Psat(comp6,B2)	=Psat(comp4,B2)
6	Tsat, I, oC	=WaterTsat(B4)	=Tsat(comp6,B4)	=Tsat(comp4,B4)
7	Relative Humidity	0.5		
8	Pi, bar	=B5*B7	=0.79*(B4-B8)	=0.21*(B4-B8)
9	zi	=B8/\$B\$4	=C8/\$B\$4	=D8/\$B\$4
10	Henry's Constants, Hi, bar		=Henry(B2,'Henry Constants'!C4:C7)	=Henry(B2,'Henry Constants'!B4:B7)
11	ki, bar	=B5/B4	=C10/B4	=D10/B4
12	xi	=1-C12-D12	=C9/C11	=D9/D11

Figure 23: Formula Audit of the Mole Fraction Calculation

As expected since air is a superheated vapor at the given relative humidity and temperature, the mole fraction of both nitrogen and oxygen are extremely small and the liquid is essentially pure water with $x_{O_2} \approx 1$.

3.2.7 Concentration of Oxygen in Pure Water

Given the liquid mole fraction of oxygen, the calculation of oxygen in water becomes a simple dimensional analysis equation to find the concentration in terms of milligrams of oxygen per liter of water. The density of the water was calculated using the Rackett function in the CHE calculator, =Rackett(compi, temp). The value of the density of water is $61.5535 \frac{kmol}{m^3}$.

$$\left(\frac{4.75 \times 10^{-6} \text{ mol } O_2}{1 \text{ mol mixture}} \right) \left(\frac{1 \text{ mol mixture}}{0.99999 \text{ mol } H_2O} \right) \left(\frac{61.5535 \text{ kmol } H_2O}{1 \text{ m}^3 \text{ } H_2O} \right) \left(\frac{1 \text{ m}^3 \text{ } H_2O}{1000 \text{ L } H_2O} \right) \left(\frac{1000 \text{ mol } H_2O}{1 \text{ kmol } H_2O} \right) \left(\frac{31999 \text{ mg } O_2}{1 \text{ mol } O_2} \right)$$

The concentration of oxygen in pure water was found to be $9.3587 \frac{mg \text{ } O_2}{L \text{ } H_2O}$. The following figures provide the excel set up and audit formula for the final calculation.

	A	B	C	D
1		Water	Nitrogen	Oxygen
2	Temperature, oC	25		
3	Temperature, K		298.15	
4	Pressure, Bar		1.01325	
5	Psat, i, bar	0.031660011	1323.499518	1761.229951
6	Tsat, I, oC	99.99983306	-195.6875406	-182.8946437
7	Relative Humidity		0.5	
8	Pi, bar	0.015830005	0.787961796	0.209458199
9	zi	0.015623001	0.77765783	0.20671917
10	Henry's Constants, Hi, bar		86367.81552	44083.38413
11	ki, bar	0.031246001	85238.40664	43506.91747
12	xi	0.999986125	9.12333E-06	4.75141E-06
13	Density, kmol/m3	61.55354604		
14	molecular weight, kg/kmol	18.015		31.999
15	1 m3 to L		1000	
16	1 g to mg		1000	
17	Concentration, mg/L			9.358752189

Figure 24: Full Excel Set Up and Results for Problem B

	A	B	C	D
1		Water	Nitrogen	Oxygen
2	Temperature, oC	25		
3	Temperature, K	=ctok(B2)		
4	Pressure, Bar	=ATMtoBAR(1)		
5	Psat, i, bar	=WaterPsat(B2)	=Psat(comp6,B2)	=Psat(comp4,B2)
6	Tsat, I, oC	=WaterTsat(B4)	=Tsat(comp6,B4)	=Tsat(comp4,B4)
7	Relative Humidity	0.5		
8	Pi, bar	=B5*B7	=0.79*(B4-B8)	=0.21*(B4-B8)
9	zi	=B8/\$B\$4	=C8/\$B\$4	=D8/\$B\$4
10	Henry's Constants, Hi, bar		=Henry(B2,'Henry Constants'!C4:C7)	=Henry(B2,'Henry Constants'!B4:B7)
11	ki, bar	=B5/B4	=C10/B4	=D10/B4
12	xi	=1-C12-D12	=C9/C11	=D9/D11
13	Density, kmol/m3	=Rackett(comp5,B2)		
14	molecular weight, kg/kmol	=molw5		=molw4
15	1 m3 to L	1000		
16	1 g to mg	1000		
17	Concentration, mg/L			=D12/B12*B13*D14*B16

Figure 25: Formula Audit for Problem B

3.3 Problem B Conclusion

Problem B demonstrated the use of partial molar properties outside of a power cycle or other process requiring thermodynamic evaluation. Thermodynamics of the partial molar properties was used to understand the solubility of components in mixtures. The greater the Henry Constant, the less solubility of a component. This is intuitive from the explanation that the constant is the pressure required to force a certain mole fraction to occur.

4 Conclusion

From sodas to sea water to shampoos to pharmaceuticals, the world and the materials that surround us are filled with mixtures. These mixtures are stable under certain conditions and the evaluation of their thermodynamic properties allows for decisions about making, storing, and selling these is needed. This assignment demonstrated the importance of the partial molar properties in regards to throttling, a common process in the refrigeration cycle. Just as in problem A, the throttling process takes a high pressure sub-cooled saturated liquid and throttles it down to a lower pressure turning it into a much colder 2-phase liquid-vapor mixture that enables the cooling effect of the evaporator. The resulting mixture and temperature must meet the required standards for efficiency requiring the evaluation of partial molar properties.

Problem B has interesting correlations to environmental aspects as well. As our planet warms from greenhouse emissions, understanding the solubility of those gases, carbon dioxide and methane especially, in the oceans and other water ways is important. Oceans are a large source of stored oxygen for our planet. Understanding how the increased carbon dioxide and other greenhouse emissions dissolve in the oceans and impact on the solubility of oxygen in the oceans as well is of the utmost importance. Thus we see that understanding thermodynamic principles extends to every part of our lives.

5 Sources

The use of Google Gemini was used to help code this assignment in overleaf.

References

- [1] “Raoult’s Law.” Chemistry LibreTexts, 2023.
- [2] “Dalton’s Law of Partial Pressure.” LabXchange, 2026.
- [3] Caracotsios, Michael. *An Introduction to Chemical Engineering Thermodynamics*. First ed., pp. 254–280.